

- [1] Incompressible means that v and u are much more sensitive to changes in T than in P .⁺¹⁰ This means you can approximate

$$v(T, P) \approx v(T) \approx v_f(T_{\text{sat}} = T)$$

$$u(T, P) \approx u(T) \approx u_f(T_{\text{sat}} = T)$$

$\Rightarrow$ applies to liquids

[2] Given: $M = 58 \text{ lb} = 26.3084 \text{ kg}$

$$L = 45 \text{ in} = 1.143 \text{ m}$$

$$L_{\text{tot}} = 20L = 22.86 \text{ m}$$

$$C = 4187 \text{ J/kg}\cdot\text{K}$$

$$M_{\text{H}_2\text{O}} = 7.84 \text{ lb} = 3.556 \text{ kg}$$

Assume: S.C.S. +1

quasi-ef. +1

a) $\Delta PE = Mg \Delta z$ ⁺² $= (26.3084)(9.8)(22.86) \rightarrow \boxed{\Delta PE = 5893.82 \text{ J}}$ ⁺¹

b) $\Delta \bar{U} = \Delta PE = M_{\text{H}_2\text{O}} c \Delta T$ ⁺³

$$\Delta T = \frac{\Delta PE}{M C} = \frac{5893.82}{(4187)(3.556)} \rightarrow \boxed{\Delta T = 0.396 \text{ K}}$$
⁺¹

c) Video: $T_{\text{initial}} = 18.152^\circ\text{C}$ ⁺¹ $= 291.302 \text{ K}$

$$T_{\text{final}} = 18.497^\circ\text{C} = 291.647 \text{ K}$$

$$\Delta T = 0.345 \text{ K}$$
⁺¹

$\Rightarrow$ differences likely due to heat loss to the colder room +1
or something reasonable

d) $T_{\text{boil}} = 100^\circ\text{C} \rightarrow \Delta T = 100 - 18.152 = 81.848^\circ\text{C}$ ⁺¹ $= \text{K}$

$\Rightarrow \Delta \bar{U} = \Delta PE \rightarrow M_{\text{weight}} g \Delta z = M_{\text{H}_2\text{O}} c \Delta T$ ⁺¹

$$\Delta z = \frac{M_{\text{H}_2\text{O}} c \Delta T}{M_{\text{weight}} g} = \frac{(3.556)(4187)(81.848)}{(26.3084)(9.8)}$$

$$\boxed{\Delta z = 4726.64 \text{ m}}$$
⁺¹ almost 5 km / $\sim 3 \text{ miles!}$

3) Given: $M = 5 \text{ kg}$ $R = 287 \text{ J/kg}\cdot\text{K}$ Assume: ideal gas +1
 $V_1 = 1 \text{ m}^3$ $C_v = 714 \text{ J/kg}\cdot\text{K}$: $C_v = \text{const.}$ +1
 $T_1 = 300 \text{ K}$: quasi- eq +1
 $T_2 = 500 \text{ K}$: process 1 = isochoric
 $V_3 = 3 \text{ m}^3$ $V_2 = V_1$ (rigid) +1
 : process 2 = isobaric
 $P_3 = P_2$
 $\Delta KE = 0$ $\Delta PE = 0$ +1

a) State 1: $P_1 V_1 = M R T_1 \rightarrow T_1 = 300 \text{ K}, V_1 = 1 \text{ m}^3$

$$P_1 = \frac{M R T_1}{V_1} = \frac{(5)(287)(300)}{1} \rightarrow \boxed{P_1 = 430500 \text{ Pa}} +1$$

State 2: $P_2 V_2 = M R T_2 \rightarrow T_2 = 500 \text{ K}$

$$\boxed{V_2 = V_1 = 1 \text{ m}^3} +1 \text{ b/c sealed}$$

$$P_2 = \frac{M R T_2}{V_2} = \frac{(5)(287)(500)}{1} \rightarrow \boxed{P_2 = 717500 \text{ Pa}} +1$$

State 3: $V_3 = 3 \text{ m}^3$

$$\boxed{P_3 = P_2 = 717500 \text{ Pa}} +1 \text{ b/c isobaric expansion}$$

$$T_3 = \frac{P_3 V_3}{M R} = \frac{(717500)(3)}{(5)(287)} \rightarrow \boxed{T_3 = 1500 \text{ K}} +1$$

b) $\Delta U = M C_v \Delta T$ +2

$$\rightarrow U_2 - U_1 = (5)(714)(500 - 300) = \boxed{714000 \text{ J}} +1$$

$$\rightarrow U_3 - U_2 = (5)(714)(1500 - 500) = \boxed{3570000 \text{ J}} +1$$

$$c) \quad {}_1\bar{W}_2 = \int_{v_1}^{v_2} P dv \quad \boxed{= 0 \text{ J}} \quad \begin{matrix} +2 \\ +1 \end{matrix} \quad b/c \quad v_1 = v_2$$

$${}_2\bar{W}_3 = \int_{v_2}^{v_3} P dv = P \int_{v_2}^{v_3} dv = P(v_3 - v_2) = 717500(3-1)$$

$$\boxed{{}_2\bar{W}_3 = 1435000 \text{ J}} \quad \begin{matrix} +2 \\ +1 \end{matrix} \quad \text{Wk is done by fluid b/c piston is pushed against surroundings during expansion} \quad +1$$

$$d) \quad \bar{U}_2 - \bar{U}_1 = {}_1Q_2 - {}_1\bar{W}_2 \rightarrow {}_1Q_2 = \Delta \bar{U}$$

$$\boxed{{}_1Q_2 = 714000 \text{ J}} \quad +1$$

$$\bar{U}_3 - \bar{U}_2 = {}_2Q_3 - {}_2\bar{W}_3$$

$${}_2Q_3 = \Delta \bar{U} + {}_2\bar{W}_3 = 3570000 + 1435000$$

$$\boxed{{}_2Q_3 = 5005000 \text{ J}} \quad +1$$

4) Given: $M = 1 \text{ kg}$

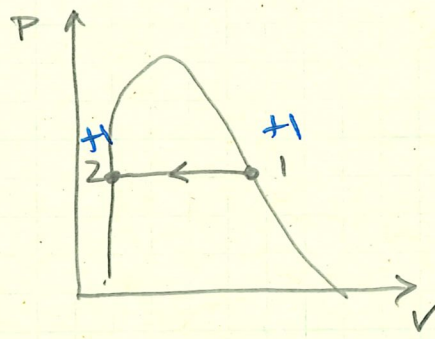
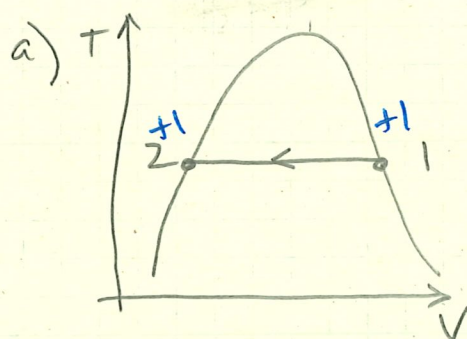
$$P_1 = P_2 = 2 \text{ MPa}$$

$$x_1 = 1, x_2 = 0$$

Assume: S.C.S., +1

: quasi-st., +1

: $\Delta KE = \Delta PE = 0$ +1



b) $T_1 = T_2 = T_{\text{sat}} = 485.53 \text{ K} \xrightarrow{+1} \Delta T = 0 \xrightarrow{+1}$ because all change in energy is due to change in phase

$$c) \quad {}_1\bar{W}_2 = \int_{v_1}^{v_2} P dv = M \int_{v_1}^{v_2} P dv = MP \int_{v_1}^{v_2} \frac{1}{v} dv = MP(v_2 - v_1) = MP(v_f - v_g)$$

$${}_1\bar{W}_2 = (1)(2000000)(0.0011767 - 0.099585)$$

$$\boxed{{}_1\bar{W}_2 = -196816.6 \text{ J}} \quad \text{work done on fluid due to compression} \rightarrow v_2 < v_1$$

$$d) \quad \Delta U = {}_1Q_2 - {}_1\bar{W}_2$$

$${}_1Q_2 = \Delta U + {}_1\bar{W}_2 = M(u_2 - u_1) + {}_1\bar{W}_2$$

$$= M(u_f - u_g) + {}_1\bar{W}_2$$

$$= (1)(906.14 - 2599.1) - 196.8166 \text{ kJ}$$

$$\boxed{{}_1Q_2 = -1889.777 \text{ kJ}} \quad \text{Heat at b/c energy is being removed from fluid, } Q < 0$$

$$\boxed{5} \quad \text{Cycle} \rightarrow \text{Assume: S.C.S., } \Delta PE = \Delta KE = 0$$

$$\text{quasi-static } {}_1Q_2 = 0, {}_3Q_4 = 0$$

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State 1 : $P_1 = 2.6 \text{ MPa}$, $T_1 = 499.2 \text{ K}$, $V_1 = 0.0012014 \frac{\text{m}^3}{\text{kg}}$
 $\Rightarrow x_1 = 0 \rightarrow u_1 = u_f = 968.55 \frac{\text{kJ}}{\text{kg}} +1$

State 2 : $P_2 = 5 \text{ MPa}$, $T_2 = 500 \text{ K}$, $V_2 = 0.0012 \frac{\text{m}^3}{\text{kg}}$
 $\Rightarrow T < T_{\text{sat}} \rightarrow \text{liquid} \rightarrow u_2 = 969.96 \frac{\text{kJ}}{\text{kg}} +1$

State 3 : $P_3 = 5 \text{ MPa}$, $T_3 = 578.5 \text{ K}$, $V_3 = 0.046114 \frac{\text{m}^3}{\text{kg}}$
 $\Rightarrow T > T_{\text{sat}} \rightarrow \text{vapor} \rightarrow u_3 = 2711.9 \frac{\text{kJ}}{\text{kg}} +1$

State 4 : $P_4 = 2.6 \text{ MPa}$, $T_4 = 499.2 \text{ K}$, $V_4 = 0.076899 \frac{\text{m}^3}{\text{kg}}$
 $\Rightarrow x_4 = 1 \rightarrow u_4 = u_g = 2602.4 \frac{\text{kJ}}{\text{kg}} +1$

Process 1-2 : $1Q_2 = 0$

$$\Delta \bar{U} = 1Q_2 - 1\bar{W}_2 \rightarrow 1\bar{W}_2 = -\Delta \bar{U} = -M(u_2 - u_1)$$

$$1\bar{W}_2 = -1.41 \text{ kJ} +1$$

Process 2-3 : $2\bar{W}_3 = \int_{V_2}^{V_3} P dV = MP_2(V_3 - V_2) = 224.57 \text{ kJ} +1$

$$\Delta \bar{U} = 2Q_3 - 2\bar{W}_3 \rightarrow 2Q_3 = \Delta \bar{U} + 2\bar{W}_3 +1$$

$$= M(u_3 - u_2) + 2\bar{W}_3 = 1966.57 \text{ kJ} +1$$

Process 3-4 : $3Q_4 = 0$

$$3\bar{W}_4 = -\Delta \bar{U} = -M(u_4 - u_3) \rightarrow 3\bar{W}_4 = 109.5 \text{ kJ} +1$$

Process 4-1 : $4\bar{W}_1 = MP_4(u_1 - V_4) = -196.814 \text{ kJ} +1$

$$4Q_1 = M(u_1 - u_4) + 4\bar{W}_1 = -1830.664 \text{ kJ} +1$$

$$\bar{W}_{\text{net}} = 1\bar{W}_2 + 2\bar{W}_3 + 3\bar{W}_4 + 4\bar{W}_1 = 135.846 \text{ kJ} +1$$

$$Q_{\text{net}} = 1Q_2 + 2Q_3 + 3Q_4 + 4Q_1 = 135.846 \text{ kJ} +1$$

$$\sum \Delta \bar{U} = Q_{\text{net}} - \bar{W}_{\text{net}} = 135.846 - 135.846 = 0 \checkmark +1$$